



Interaksi Manusia dan Komputer (IMK) THE USER INTERFACE (UI) DESIGN PROCESS

Step 6

Select the proper interaction device

Step – 7

Screen-Based Control

Interaksi Manusia dan Komputer (IMK)

THE USER INTERFACE (UI) DESIGN PROCESS

Step 6

Select the proper interaction device



Tujuan

- Setelah mengikuti materi ini mahasiswa dapat:
Mahasiswa/i dapat mampu memahami karakteristik setiap perangkat interaksi sehingga dapat memilih perangkat interaksi yang tepat diterapkan dalam desain UI.

Interaction Device



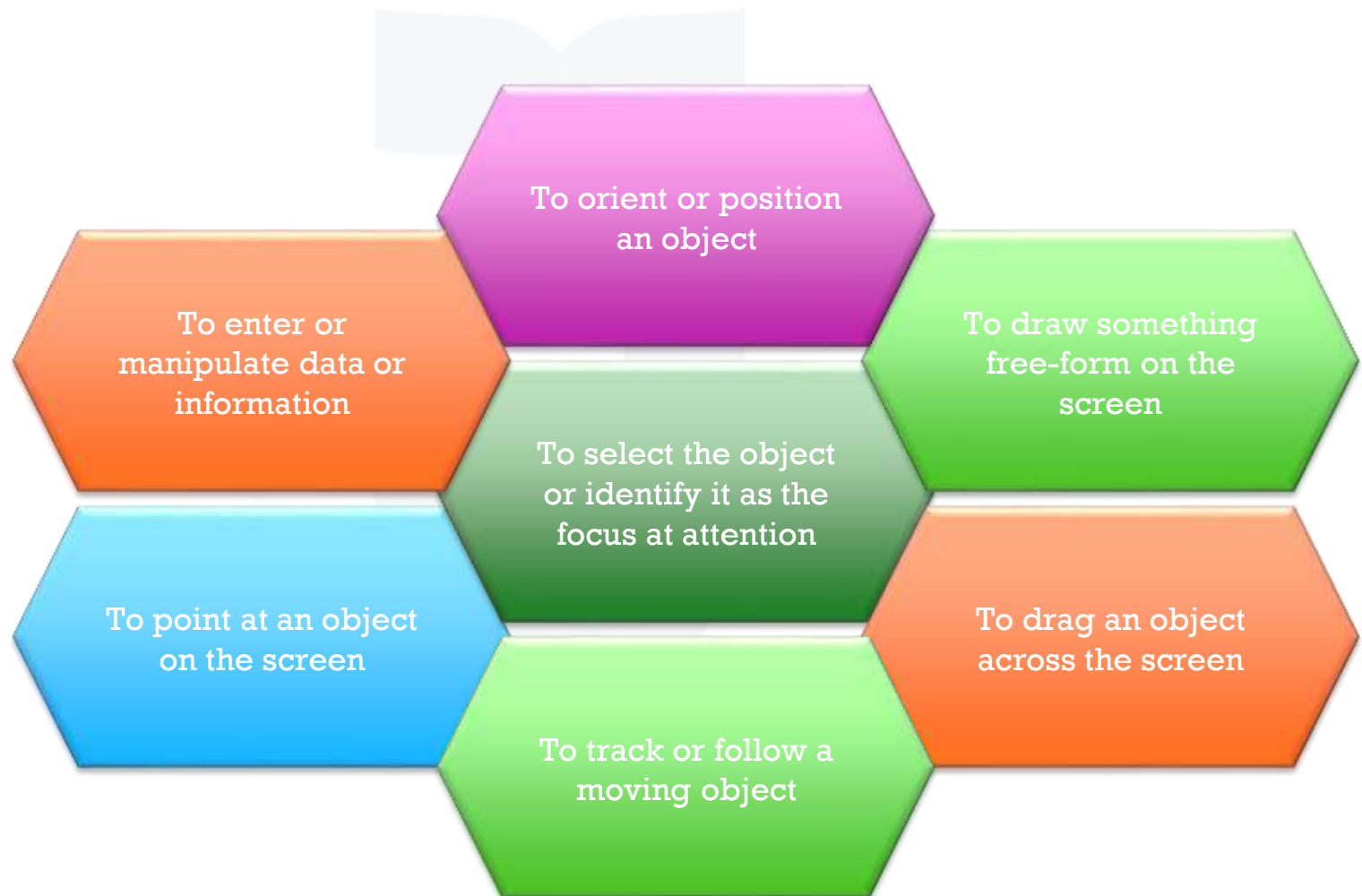
Adalah suatu mekanisme input atau device yang digunakan user untuk mengkomunikasikan keinginan dan kebutuhannya kepada komputer, dan/atau mekanisme Output atau device yang digunakan komputer untuk memberikan respond kepada user

PEMILIHAN DEVICE

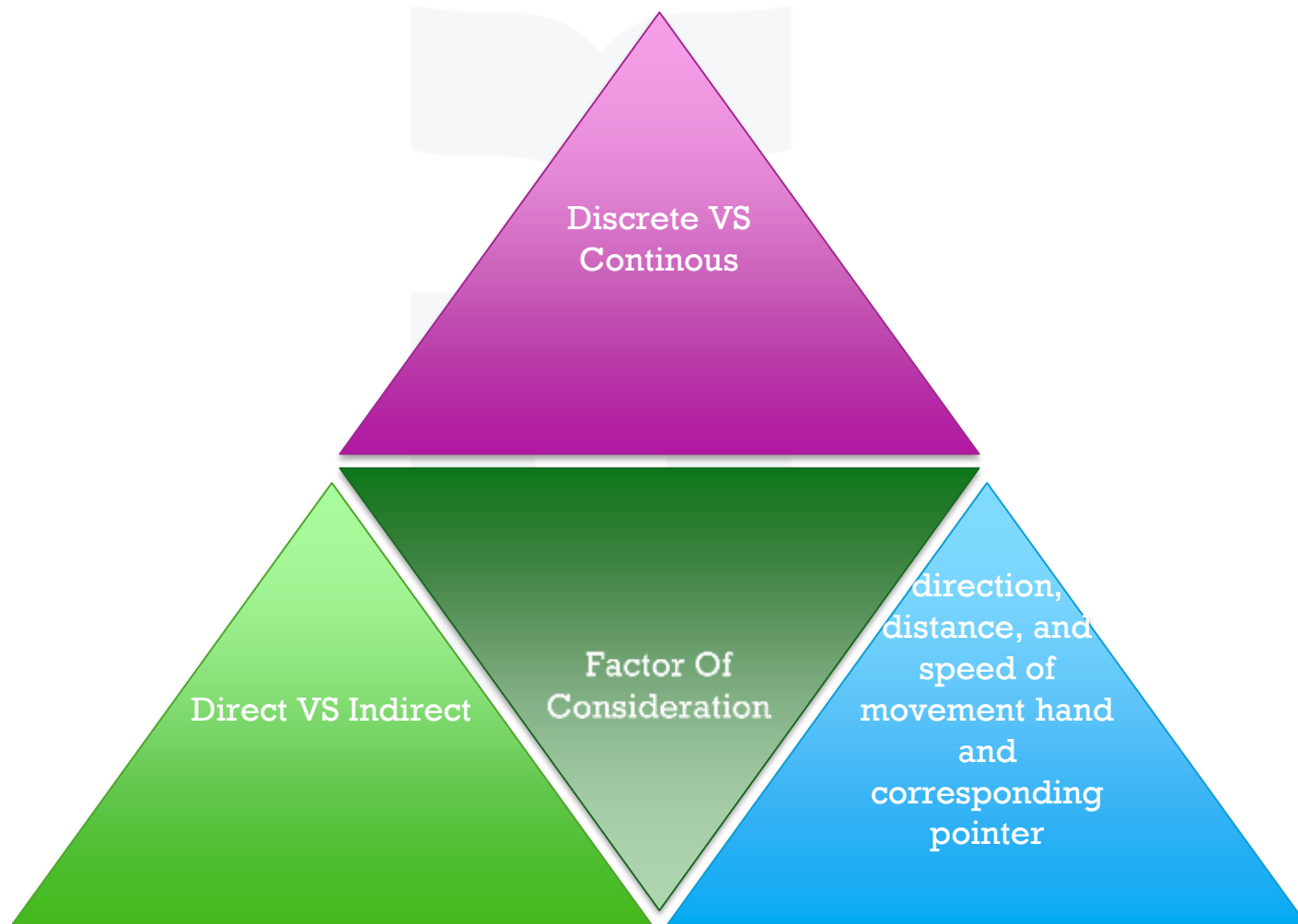
- Pemilihan Device yang tepat adalah hal **PENTING**
- Jika tepat maka Performansi Cepat dan akurat. Jika tidak tepat produktivitas turun error naik, **STRESS**



CHARACTERISTICS OF INPUT DEVICE



FACTOR OF CONSIDERATION



Input Device

- Mouse
- Trackball
- Joystick
- Touch screen
- Graphic Tablet or Trackpad
- Keyboard
- Voice
- Gesture Recognition
- Facial Expression
- Eye Tracking device
- Fingerprint
- Handwriting

SELECTING THE PROPER INPUT DEVICE



Panduan dalam memilih Input device yang sesuai

- ☐ Karakteristik Task
- ☐ Karakteristik User dan preferences
- ☐ Karakteristik dari environment
- ☐ Karakteristik dari Hardware
- ☐ Karakteristik dari device dipandang dari aplikasi yang dibuat/digunakan
- ☐ Fleksibilitas
- ☐ Minimalisir pergerakan tangan dan mata

Pointer Guidelines

- ❑ Pointer :
 - ➔ Harus selalu terlihat
 - ➔ Harus kontras dengan background
 - ➔ Harus memiliki ukuran yang konsisten ketika bergerak
 - ➔ Area Hotspot harus mudah dikenali posisinya
- ❑ User harus selalu bisa menentukan posisi pointer
- ❑ Bentuk pointer
- ❑ Pergunakan bentuk/variasi bentuk dari pointer secukupnya
- ❑ Berikan kebebasan kepada user saat menggerakkan pointer di dalam layar
- ❑ Animasi seharusnya tidak:
 - ➔ membingungkan
 - ➔ Mengurangi kemampuan untuk berinteraksi

Output Device

- ❑ Screen
 - ➔ Image
 - ➔ Color
 - ➔ Size
- ❑ Speaker

Penggunaan Sound bisa berupa simple beep, speech, music dan sound effect.

CHALLENGES

- Analisis kelebihan dan kekurangan masing-masing device di bawah ini:
 - Trackpad Macbook
 - Trackpad Windows
 - Trackball Blackberry
 - Trackpoint Thinkpad

Interaksi Manusia dan Komputer (IMK)

THE USER INTERFACE (UI) DESIGN PROCESS

Step – 7 :
Screen-Based Control



Tujuan

- Setelah mengikuti materi ini mahasiswa dapat:
Mampu memilih dan menerapkan *screen-based controls* yang tepat dengan kebutuhan

Screen Control

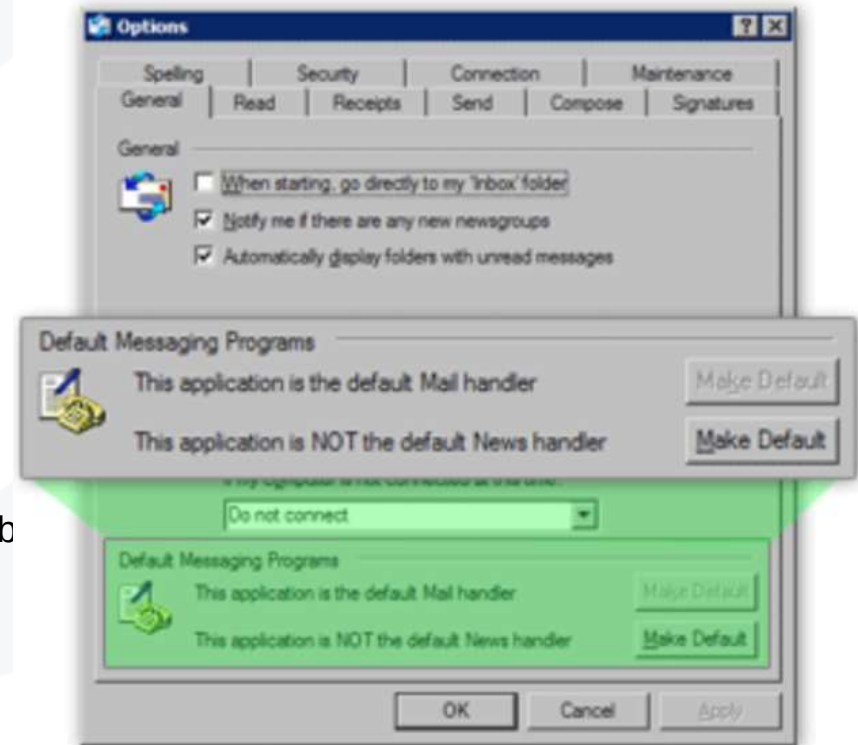
- Sometimes called ‘widgets’
- Widgets: the elements of a screen that constitute its body.
- Definition: graphic objects that represent the properties or operations of other objects
- Functions:
 - Permit the entry or selection of a particular value
 - Permit the changing or editing of a particular value
 - Display only a particular piece of text, value, or graphic
 - Cause a command to be performed
 - Possess a contextual pop-up window

Three Extremely Important Principles

- A control must:
 - Look the way it works
 - Work the way it looks
- “Look” → “enterability” or “clickability”
- Used exactly as its design intended
- Presented in a standard manner

Example:

- Windows XP:
 - Raised elements can be pressed
 - Recessed elements can not be pressed
 - Elements on a flat white background can b



Fact about Web page design

- Unleashed and exposed thousands of instances
- Basic principles have been violated
- Placing greater value on personal creativity than on interface usability

The screenshot shows the Garuda Search flight booking interface. At the top, there are three tabs: 'Cari Penerbangan' (selected), 'Flight + Hotel', and 'Garuda Search'. Below the tabs is a green-bordered form with the following fields:

- Dari** (Origin): A text input field with a search icon on the right.
- Ke** (Destination): A text input field with a search icon on the right.
- One-way or Round-trip**: Two radio buttons, 'Satu arah' (selected) and 'Kembali'.
- Berangkat** (Departure): A date picker field.
- Kembali** (Return): A date picker field.
- 1** (Passengers): A text input field with a person icon on the right.
- Anak-anak** (Children): A text input field with a person icon on the right.
- Bayi** (Infants): A text input field with a person icon on the right.
- Rupiah (IDR)** (Currency): A text input field with a currency icon on the right.

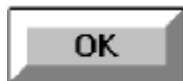
At the bottom of the form is a grey button labeled 'Cari Penerbangan' with a search icon.

Operable Controls

- ... are those that permit:
 - Entry
 - Selection
 - Changing
 - Editing
 - Cause a command to be performed
- Here are the classes... (next slide)

1 – Operable Buttons

- Description
 - A square or rectangular-shaped control with a label inside that indicates action to be accomplished
 - The label may consist of text, graphics, or both
- Purpose
 - To start actions
 - To change properties
 - To display a pop-up menu
- There are two kinds:



Command buttons



Toolbar buttons

1 – Operable Buttons

- **Advantages**

- Always visible, reminding one of the choices available
- Convenient
- Can be logically organized in the work area
- Can provide meaningful descriptions of the actions that will be performed
- Larger size generally provides faster selection target

- **Disadvantages**

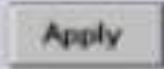
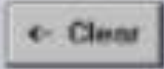
- Consumes screen space
- Size limits the number that may be displayed
- Requires looking away from main working area to activate
- Requires moving the pointer to select

- Can possess 3-D appearance
 - Adds an aesthetically pleasing style to the screen
 - Provides visual feedback through button movement when activated
- May permit use of keyboard equivalents and accelerators
- Faster than using a two-step menu bar/pull-down sequence

1 – Operable Buttons

- To a window:
 - To cause something to happen immediately
 - To display another window
 - To display a menu of options
 - To set a mode or property value
- In a web
 - Use links to show information
- Things to consider:
 - Usage
 - Structure
 - Labels
 - Size
 - Number
 - Location & layout
 - Organization
 - Intent indicators
 - Expansion buttons
 - Default
 - Keyboard equivalent / accelerators
 - Button activation

1 – Operable Buttons

<i>Intent Indicator</i>	Contoh
Melaksanakan aksi langsung	
Membuka <i>cascading dialog</i>	
Membuka menu	
Membuka <i>expanding dialog</i>	
Memiliki ketergantungan dengan kontrol lain	

2 – Text Entry / Read only controls

- Usually referred as “*field*” or “*text-box*”
 - *Unprotected*: a text box into which information can be keyed
 - *Protected*: A text box used for display purposes only
- Description
 - Usually rectangular in shape
 - Usually possesses a caption or label describing the kind of information contained within it
 - Two types exist
 - Single-line
 - Multiple-line
 - When first displayed, the box may be blank or contain an initial value
- Purpose
 - To permit the display, entering, or editing of textual information
 - To display read-only information

Single occurrence

Composition:

Or

Composition:

Multiple occurrence

Offices:

Read-only / display

Location:

Alice Springs
Kakadu National Park
Traralgon
Wagga Wagga
Whyalla

If alphanumeric then
left justify

Balances:

12,642,123.05
53.98
355,125.44
199.13
612.01

If numeric then right
justify

Segmentation

Date:

Telephone:

Or

Date: / /

Telephone: () -

2 – Text Entry / Read only controls

- **Advantages**
 - Very flexible
 - Familiar
 - Consumes little screen space
- **Disadvantages**
 - Requires use of typewriter keyboard.
 - Requires user to remember what must be keyed
- **Proper usage**
 - Most useful for data that is
 - Unlimited in scope
 - Difficult to categorize
 - Of a variety of different lengths
 - When using a selection list is not possible

3 – Selection Controls

- Presents on the screen all the possible alternatives, conditions, or choices that may exist for an entity, property, or value
- Types
 - Radio button
 - Check box
 - Palette
 - List box
 - List view control
 - Drop-down / Pop-up list box

The diagram shows two examples of selection controls. On the left is a 'palette' consisting of a rectangular box containing four radio buttons, each followed by a color name: Red, Yellow, Green, and Blue. To the right of this box is the word 'Or'. To the right of 'Or' is a 'list box', which is a horizontal rectangular box containing four radio buttons, each followed by a color name: Green, Blue, Yellow, and Red.

3 – Selection Controls: Radio button

“Only one selection permitted”

Poor

Plan Choice: ☐ Limited ☐ Basic ☐ Superior ☐ Premium

Poor

Plan Choice: Limited Basic Superior Premium

Poor

Plan Choice: ☐ Limited ☐ Basic ☐ Superior ☐ Premium

Poor

Plan Choice: Limited ☐ Basic ☐ Superior ☐ Premium ☐

Better

Plan Choice: ☐ Limited ☐ Basic ☐ Superior ☐ Premium

3 – Selection Controls: Radio button

Still better

Plan Choice:

☐ Limited
☐ Basic
☐ Superior
☐ Premium

Plan Choice:

☐ Limited
☐ Basic
☐ Superior
☐ Premium

Plan Choice:

☐ Limited ☐ Basic ☐ Superior ☐ Premium

Best !

Plan Choice:

☐ Limited
☐ Basic
☐ Superior
☐ Premium

Plan Choice

☐ Limited
☐ Basic
☐ Superior
☐ Premium

Color

☐ Green ☐ Blue ☐ Yellow ☒ Red

3 – Selection Controls: Check box

“Use for multiple selection”

- ☒ Bold
- ☐ Italic
- ☐ Subscript
- ☒ Underline

- ☒ Bold
- ☐ Italic
- ☐ Underline

- ☐ Always Create Backup Copy
- ☒ Allow Fast Saves
- ☐ Prompt for Document Properties
- ☐ Prompt to Save Normal Template
- ☐ Save Native Picture Formats Only
- ☐ Embed TrueType Fonts
- ☐ Save Data Only for Forms
- ☒ Automatic Save Every:

Earnings:

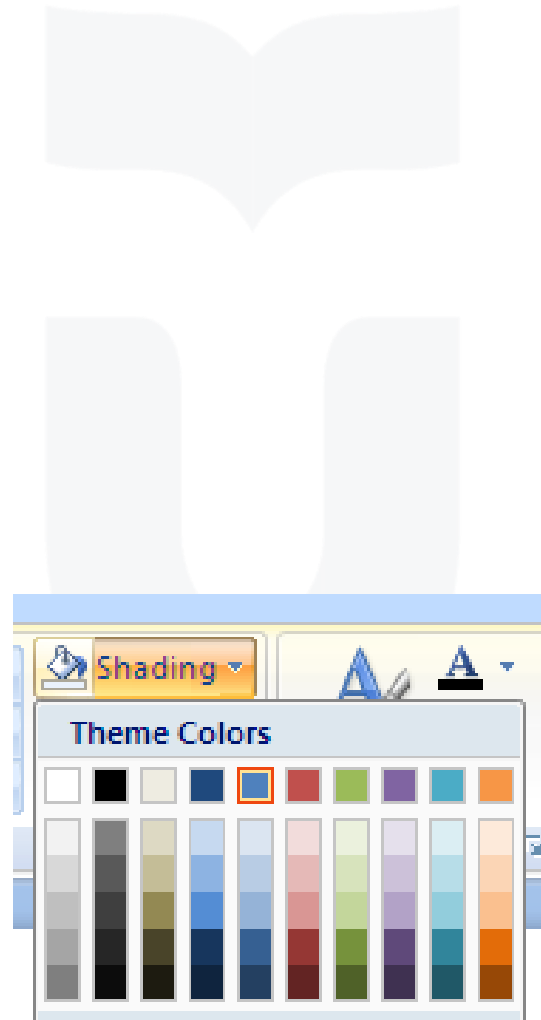
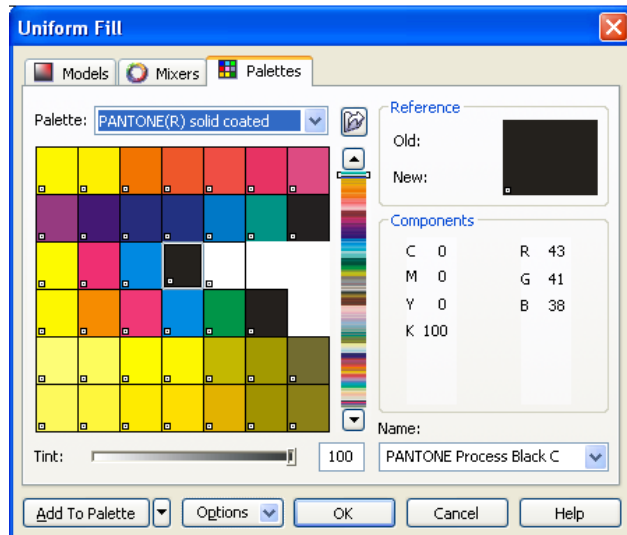
- ☐ Annual
- ☐ Quarterly
- ☐ Monthly
- ☐ Weekly

Earnings

- ☐ Annual
- ☐ Quarterly
- ☐ Monthly
- ☐ Weekly

Best !

3 – Selection Controls: Palette



3 – Selection Controls: List box

Events:

- Asterisk
- Chat Incoming Ring
- Chat Outgoing Ring
- Critical Stop
- Default Beep
- Exclamation
- Money Transaction Entry
- Question

Files:

- <none>
- chimes.wav
- chord.wav
- cointabl.wav
- ding.wav
- kerchunk.wav
- ringin.wav
- ringout.wav

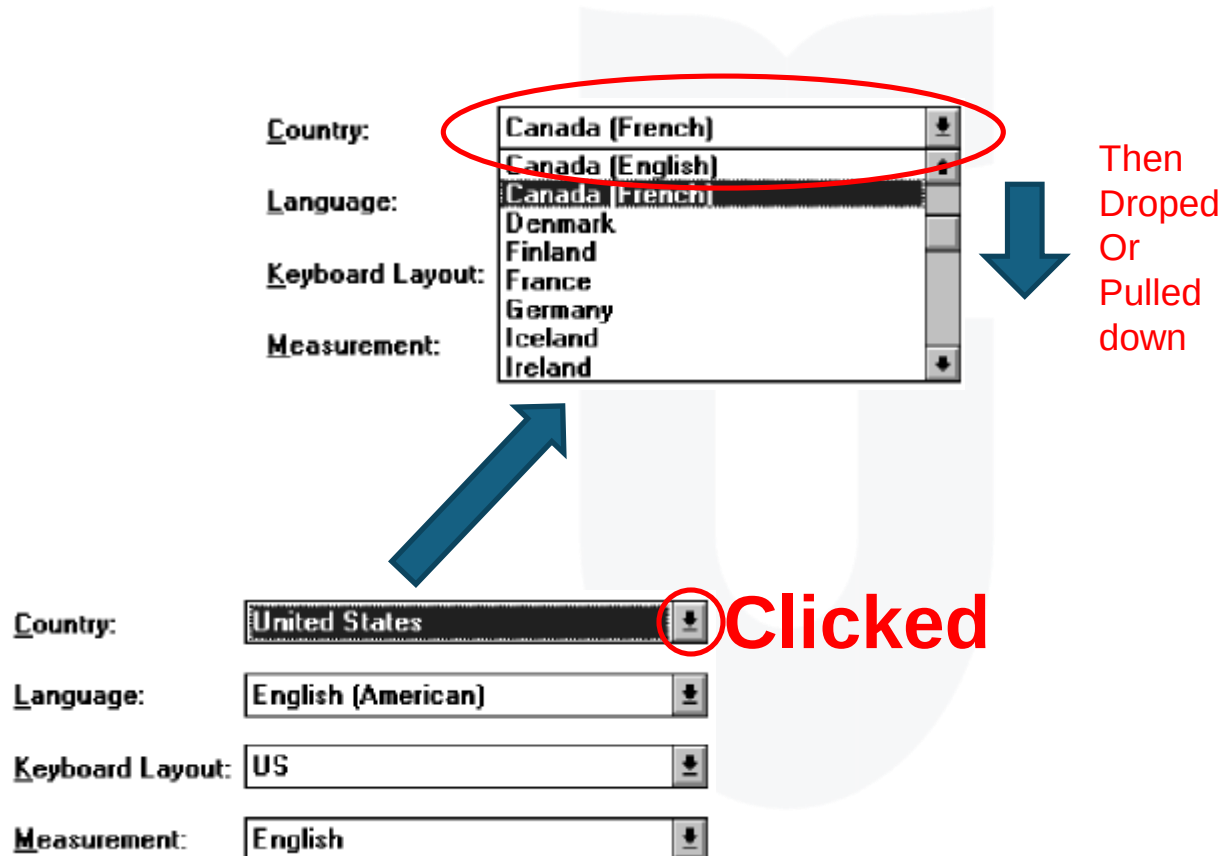
List box with multiple selection

Groceries:

4 selected

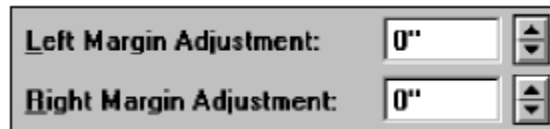
- ☒ Bread
- ☐ Cereal
- ☒ Dairy Foods
- ☐ Desserts
- ☐ Drinks
- ☐ Fruit
- ☒ Meat, Fish and Poultry
- ☐ Vegetables

3 – Selection Controls: Drop Down/Pull-down



Combination Entry/Selection Controls

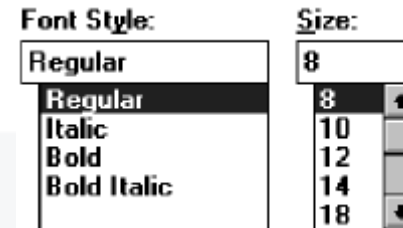
Spin boxes



Left Margin Adjustment: 0" 

Right Margin Adjustment: 0" 

Combo boxes



Font Style: Regular
Regular
Italic
Bold
Bold Italic

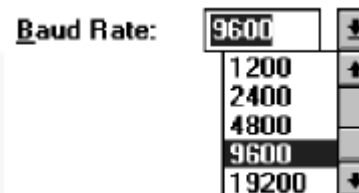
Size: 8
8
10
12
14
18

Combo box - Closed



Baud Rate: 9600 

Combo box - Opened

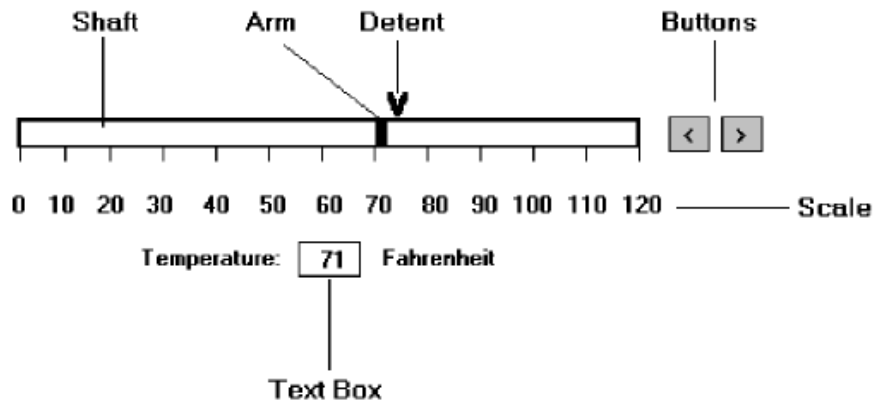


Baud Rate: 9600 

1200
2400
4800
9600
19200

Other Operable Controls

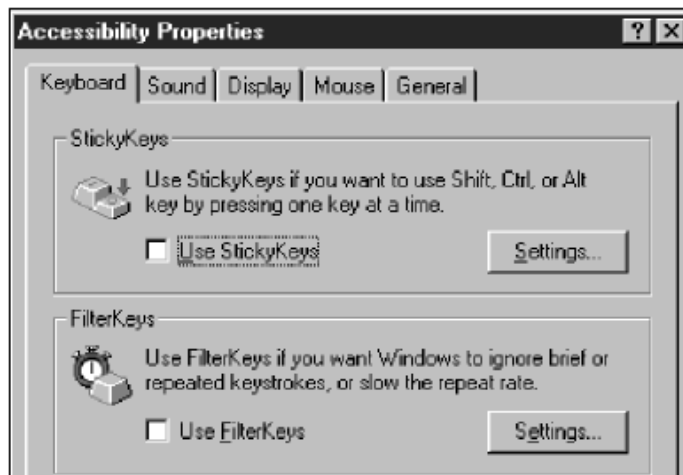
Slider



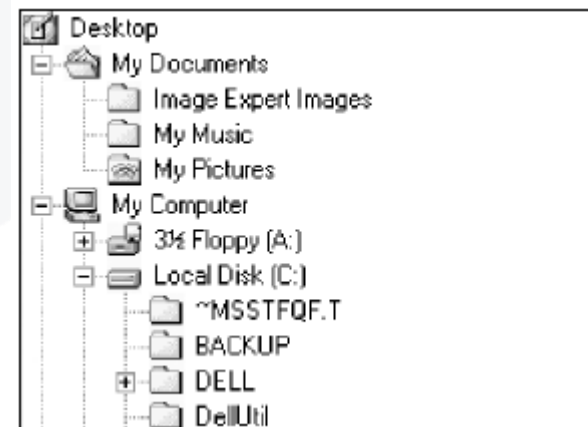
Date picker



Tabs



Tree view



Custom Controls

- Implement custom controls with caution
- The addition of custom controls adds to this learning and increases system complexity



Presentation Controls

- Static text fields: read-only textual information
- Group box: a rectangular frame that surrounds a control or group of controls; optional caption may included.
- Column headings: read-only textual information that serves as a heading above columns of text or numbers
- Tool tip/balloon tip: a small pop-up window that contains information; could be placed in balloon.

Caption:
HEADING
This message is very important!

Static text

Group Box

Optional caption

Column heading

Name	Size	Type	Modified
 11-12.bmp	233 KB	Bitmap Image	1/23/95 3:00 PM
 11-13.bmp	470 KB	Bitmap Image	1/23/95 3:01 PM
 11-14.bmp	151 KB	Bitmap Image	1/17/95 5:05 PM
 11-15.bmp	151 KB	Bitmap Image	1/17/95 5:06 PM

Column part

Tool tip / balloon tip



Time

78

You have entered an invalid character.
Valid characters are 0 thru 9, *, and #.

Progress indicators

Updating...

80% Complete



Stop

Selecting the Proper Controls

- Choose familiar controls
- Consider the task
- Reduce the number of “clicks”
- Display as many control choices as possible
- When to permit text entry?
 - Permit text entry if any of the following questions can be answered “**Yes**”
 - Is the data unlimited in size and scope?
 - Is the data familiar?
 - Is the data not conducive to typing errors?
 - Will typing be faster than choice selection?
 - Is the user an experienced typist?

Selecting the Proper Controls

- Buttons or Menus for commands?
 - Consider the following:
 - Whether or not the command is part of a standard tool set.
 - The total number of commands in the application.
 - The complexity of the commands.
 - The frequency with which commands are used.
 - Whether or not the command is used in association with another control.